Classifying Galactic Center Sources Through X-ray Time-Domain Analysis

Meredith LeFevre[1], Dustin Swarm[1]

[1]Department of Physics and Astronomy, University of Iowa

Department of Physics and Astronomy

Introduction

- A star's initial mass determines its evolution and final stage. Low-mass stars (0.4-4 $M_{\odot}$) end their lives as white dwarfs (WDs), the hot, dense cores left behind after shedding their outer layers.
- Cataclysmic Variables (CVs) are close binary systems comprised of a white dwarf and a less evolved companion star where the WD pulls material from its companion, which heats to extreme temperatures as it accretes, emitting X-rays.
- The Galactic Center (GC) contains thousands of X-ray point sources with spectra consistent with CVs and the old, low-mass population of the GC makes an ideal environment to study WDs and CVs.

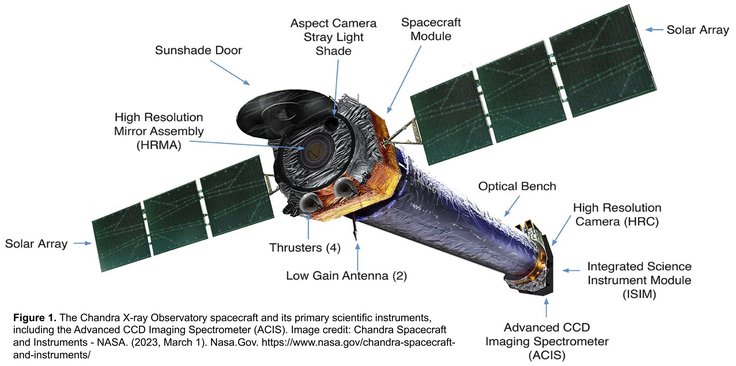


Figure 1. The Chandra X-ray Observatory spacecraft and its primary scientific instruments, including the Advanced CCD Imaging Spectrometer (ACIS). Image credit: Chandra Spacecraft and Instruments - NASA. (2023, March 1). Nasa.Gov. https://www.nasa.gov/chandra-spacecraft-and-instruments/

- Chandra X-Ray Observatory (CXO) is a NASA space telescope whose high angular resolution allows individual X-ray sources to be studied in crowded regions like the GC.
- The Advanced CCD Imaging Spectrometer (ACIS) designed to detect X-rays and measure their energies.

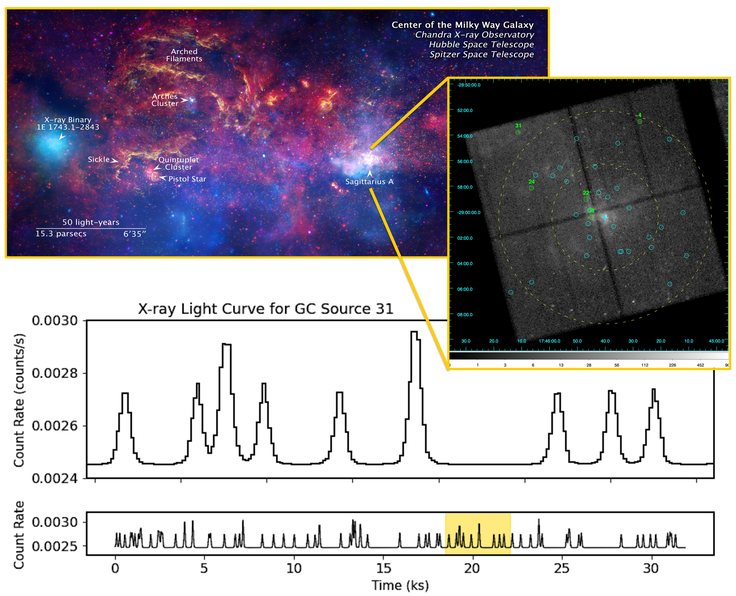
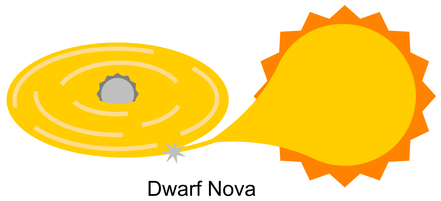
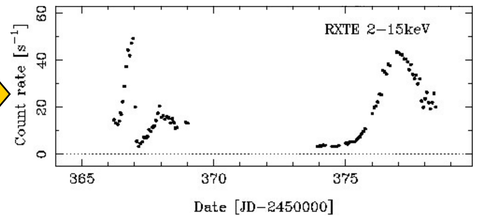
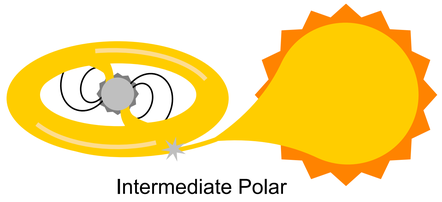


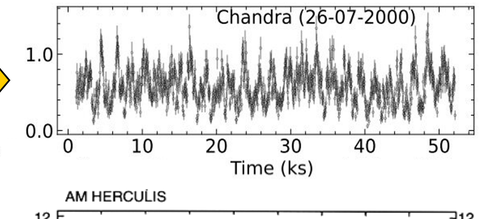
Figure 2. (Top) Chandra X-ray image of the Galactic Center. Image credit: More Images of Galactic Center (2022). NASA/CXC. https://chandra.si.edu/photo/2009/galactic/more.html. (Middle) Galactic Center image showing identified X-ray point sources (circled), with the Chandra angular aperture overlaid. Image credit: Swarm & DeRoo (in prep.). (Bottom) Chandra ACIS X-ray light curve of Galactic Center source 31 from observation 13508, obtained from the Chandra Source Catalog. The light curve was processed with the Gregory–Loredo Bayesian algorithm to identify variability.

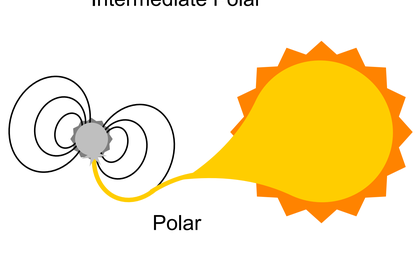
Types of Cataclysmic Variables











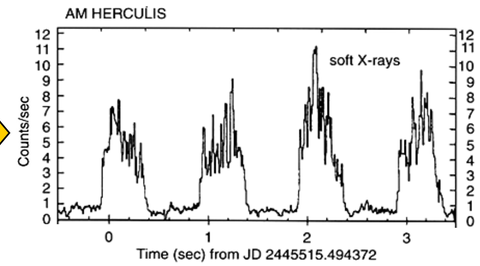


Figure 3. Schematic diagrams of the three main types of cataclysmic variables, arranged by magnetic field strength: dwarf nova (top), intermediate polar (middle), and polar (bottom)

Figure 4. (Top) Rossi X-ray Timing Explorer (RXTE) light curve of the dwarf nova SS Cygni during an outburst. (Middle) X-ray light curves of the intermediate polar TX Col. (Bottom) Soft X-ray light curve of the polar AM Herculis (AM Her). Top and bottom images reproduced from Cataclysmic Variables (NASA, 2004); middle image reproduced from Pandey et al. (2023).

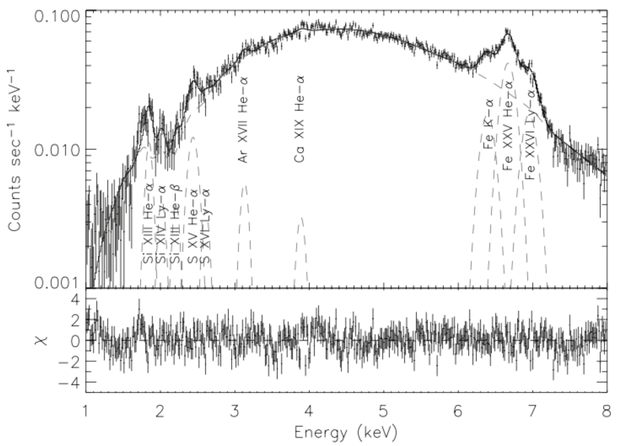


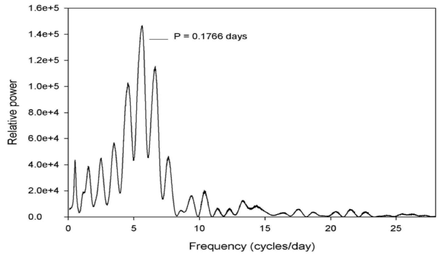
Figure 5. Combined, background-subtracted X-ray spectrum of Galactic Center point sources with fewer than 500 net counts. The overlaid best-fit model consists of a power law and 10 Gaussian emission lines, with the expected centroid energies of Si, S, Ar, Ca, and Fe indicated. Reproduced from Muno et al. (2004).

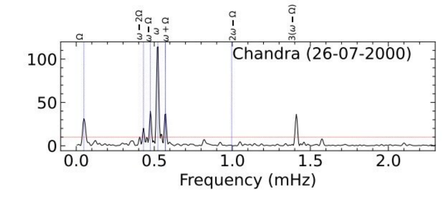
- CVs produce similar X-ray spectra, making classification difficult and potentially leading to inaccurate mass estimates.
- Consequently, many GC WDs appear unusually low-mass, challenging current models of stellar and Galactic evolution.
- X-ray emission and variability differences help distinguish CV subtypes and constrain WD masses.

- X-ray thermal spectra distinguish CVs from other accreting compact objects, particularly through iron emission lines.
- Iron emission lines trace plasma conditions and accretion geometry.
- The 6.4 keV Fe Kα line provides a useful diagnostic of CV subtypes.
- Differences in WD magnetic field strength determine accretion geometry between the difference types of CVs.
- WD mass estimates depend on the temperature of the accreted gas, which varies with accretion geometry.

$$kT_s = \frac{3}{8}\frac{G\mu M m_H}{R}$$

Methods and Discussion





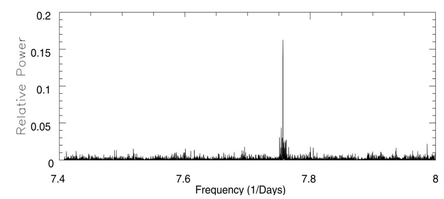


Figure 6. (Top) Lomb–Scargle periodogram of the Hα radial velocities of ES Dra. The highest peak, corresponding to the likely orbital period of 0.1766 days (4.24 hours), is marked. Reproduced from Ringwald and Velasco (2012). (Middle) Power spectra of the intermediate polar TX Col. Reproduced from Pandey et al. (2023). (Bottom) Periodogram of the polar AM Herculis (AM Her). Reproduced from Kafka et al. (2005).

- X-ray light curves track brightness changes over time, revealing periodic signals associated with physical processes in the system.
- In polars, the WD spin is synchronized with the binary orbit, producing variability at the orbital period.
- In intermediate polars, the WD spin and orbital periods are distinct, potentially producing variability at both periods.
- In dwarf novae, accretion occurs through a disk, and variability can be associated with the orbital period. [3]
- Identifying periodic signals helps distinguish CV types, constrain accretion geometries, improve WD mass estimates, and characterize the GC X-ray source population.

Future Work

- Incorporating periodicity search algorithms into a software pipeline for light curve analysis.
- Expand the X-ray time-domain analysis to all 34 selected Galactic Center sources.
- Search for periodic variability associated with WD spin and binary orbital periods to distinguish IP and non-magnetic CVs.
- Compare light curve characteristics across sources to better constrain their accretion geometries.
- Use CV classifications to refine WD mass estimates and investigate the unusually low masses inferred for Galactic Center sources.

References and Acknowledgments

1. Cataclysmic variables. (2004, March 22). Nasa.Gov. https://heasarc.gsfc.nasa.gov/docs/objects/cvs/cvs.html
2. Kafka, S., Honeycutt, R. K., Howell, S. B., & Harrison, T. E. (2005). A multiwavelength study of AM Herculis during the 2002–2004 low states. The Astronomical Journal, 130(6), 2852–2865. https://doi.org/10.1086/497893
3. Muno, M. P., Arabadjis, J. S., Baganoff, F. K., Bautz, M. W., Brandt, W. N., Broos, P. S., Feigelson, E. D., Garmire, G. P., Morris, M. R., & Ricker, G. R. (2004). The spectra and variability of X-ray sources in a Deep Chandra Observation of the galactic center. The Astrophysical Journal, 613(2), 1179–1201. https://doi.org/10.1086/423164
4. Pandey, J. C., Rawat, N., Rao, S. M., Joshi, A., & Singh, S. (2023). X-ray observations of the intermediate polar TX Col. arXiv. https://doi.org/10.48550/arXiv.2309.03674.
5. Ringwald, F. A., & Velasco, K. (2012). The orbital period and variability of the dwarf nova ES Draconis. New Astronomy, 17(2), 108–111. https://doi.org/10.1016/j.newast.2011.07.007

I would like to thank Dr. Dustin Swarm for his guidance and support throughout this project, as well as the Office for Undergraduate Research for providing funding. Research reported in this presentation is supported by the University of Iowa (UI) Office of Undergraduate Research, the Office of the Vice President for Research, the Office of the Provost, and the Sally and Ken Mason Student Success Fund.